

```
CREATE TABLE public.charity (  
    charity_id bigint GENERATED ALWAYS AS IDENTITY NOT NULL,  
    first_name character varying NOT NULL,  
    last_name character varying NOT NULL,  
    email character varying NOT NULL UNIQUE,  
    description text,  
    verified_status boolean NOT NULL DEFAULT false (verified , pending, rejected),  
    created_at timestamp with time zone NOT NULL DEFAULT now(),  
    password character varying NOT NULL,  
    rejection_reason character varying (should contain letters and numbers),  
    phone_numbers text[], -- Multi-valued: array of phone numbers  
    CONSTRAINT charity_pkey PRIMARY KEY (charity_id)  
);
```

-- Admin table

```
CREATE TABLE public.admin (  
    admin_id bigint GENERATED ALWAYS AS IDENTITY NOT NULL,  
    first_name character varying NOT NULL,  
    last_name character varying NOT NULL,  
    email character varying NOT NULL UNIQUE,  
    password character varying NOT NULL,  
    CONSTRAINT admin_pkey PRIMARY KEY (admin_id)  
);
```

-- Donor table

```
CREATE TABLE public.donor (  
    donor_id bigint GENERATED ALWAYS AS IDENTITY NOT NULL,  
    first_name character varying NOT NULL,  
    last_name character varying NOT NULL,  
    email character varying NOT NULL UNIQUE,  
    phone_numbers text[], -- Multi-valued: array of phone numbers
```

```
created_at timestamp without time zone NOT NULL DEFAULT CURRENT_TIMESTAMP,  
password character varying NOT NULL,  
CONSTRAINT donor_pkey PRIMARY KEY (donor_id)  
);
```

-- Campaign table (depends on charity)

```
CREATE TABLE public.campaign (  
    campaign_id bigint GENERATED ALWAYS AS IDENTITY NOT NULL,  
    charity_id integer NOT NULL,  
    title character varying NOT NULL,  
    description text NOT NULL (should contain letters and numbers),  
    target_amount real NOT NULL,  
    current_amount real NOT NULL DEFAULT 0,  
    start_date date NOT NULL DEFAULT CURRENT_DATE,  
    end_date date NOT NULL,  
    category character varying NOT NULL,  
    status character varying NOT NULL DEFAULT 'Pending' (Can be pending, rejected, verified),  
    rejection_reason character varying (should contain letters and numbers),  
    tags text[], -- Multi-valued: campaign tags/categories  
    location_info jsonb, -- Composite: {country, city, address}  
    CONSTRAINT campaign_pkey PRIMARY KEY (campaign_id),  
    CONSTRAINT campaign_charity_id_fkey FOREIGN KEY (charity_id) REFERENCES  
public.charity(charity_id),  
    CONSTRAINT campaign_title_charity_id_unique UNIQUE (title, charity_id) -- Composite unique  
);
```

-- Donation table (depends on donor and campaign)

```
CREATE TABLE public.donation (  
    donation_id bigint GENERATED ALWAYS AS IDENTITY NOT NULL,  
    donor_id integer NOT NULL,  
    campaign_id integer NOT NULL,
```

```
amount double precision NOT NULL,  
donation_date timestamp with time zone NOT NULL DEFAULT now(),  
payment_method character varying NOT NULL,  
transaction_status character varying NOT NULL,  
transaction_reference character varying NOT NULL UNIQUE,  
payment_details jsonb, -- Composite: {card_last4, bank_name, etc.}  
CONSTRAINT donation_pkey PRIMARY KEY (donation_id),  
CONSTRAINT donation_donor_id_fkey FOREIGN KEY (donor_id) REFERENCES  
public.donor(donor_id),  
CONSTRAINT donation_campaign_id_fkey FOREIGN KEY (campaign_id) REFERENCES  
public.campaign(campaign_id)  
);
```